

Team AI-Agent Master Prompts

Automated Threat Modeling & Attack Path Mapper — 12-Day MVP

These are ready-to-paste prompts for Member 2 and Member 3's AI coding assistants (Antigravity, Copilot, Cursor, Claude Code, etc.). Each one is scoped to *only* that member's ownership area, references the shared Pydantic contract as the interface, and explicitly lists what the AI must refuse to build.

HOW TO USE THESE

1. Give each teammate only their own prompt.
 2. Everyone's AI reads/uses `app/models/schemas.py` as the shared contract — nobody edits it without team agreement.
 3. If an AI tool ever proposes touching a file outside its owned folders, that's a signal to stop and flag it to the team, not to proceed.
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MEMBER 2 — DISCOVERY + BACKEND (≈25–30%)

PROJECT: Automated Threat Modeling & Attack Path Mapper (12-day MVP)

YOUR ROLE: Discovery + Backend Engineer (~25-30% of the project)

MISSION

You build the "eyes and nervous system" of the product: given an authorized target URL, safely and non-destructively discover publicly observable information about it, and orchestrate the pipeline that moves that data to the security-analysis engine and back to the frontend.

YOU OWN THESE FOLDERS ONLY:

- `app/discovery/`
- `backend/`

YOU MUST NOT TOUCH:

- `app/security/` (owned by AI+Cybersecurity lead)
- `app/rag/` (owned by AI+Cybersecurity lead)

- app/ai/ (owned by AI+Cybersecurity lead)
- knowledge_base/ (owned by AI+Cybersecurity lead)
- frontend/ (owned by Frontend/Platform member)
- app/models/schemas.py – READ ONLY. This is the shared contract. If you believe it needs a new field, propose it to the team; do not edit it unilaterally.

SCOPE BOUNDARY – READ THIS CAREFULLY

This is an AUTHORIZED, NON-DESTRUCTIVE discovery tool, not a scanner, fuzzer, exploit tool, or penetration-testing system. You may only:

- Perform passive/light-touch HTTP requests to pages that are part of the normal public browsing surface of the authorized target (GET requests, following visible links/forms).
- Read response headers, page content, and metadata that any normal browser visit would retrieve.

You must NOT:

- Attempt authentication bypass, brute force, SQLi/XSS payloads, fuzzing, parameter tampering, or any request designed to trigger a vulnerability.
- Hit rate limits aggressively – implement politeness delays and respect robots.txt and any explicit out-of-scope paths the team defines.
- Scrape or store personal data beyond what's needed to note that a field exists (e.g., note "password field present," never capture real values).

A missing security header or the presence of a login form is an OBSERVATION, not proof of a vulnerability – do not editorialize about security implications in your output; that's the cybersecurity layer's job. Your output should be neutral, structured fact-reporting only.

WHAT YOU BUILD

1. URL handling & validation

- Validate the submitted URL is well-formed, http(s), and within the agreed authorization scope (domain allow-list defined by the team).
- Create an analysis-job record with a unique ID and status (pending/running/completed/failed).

2. Discovery engine (app/discovery/)

- Basic HTTP/page fetch: status code, headers, title, content-type.
- Crawl publicly linked pages within the same domain up to a configurable depth/page limit (politeness delay between requests).
- Extract: links, forms (with field names/types, never real submitted values), visible API/endpoint indicators (e.g., fetch calls or hardcoded paths visible in page source/JS bundles).
- Technology fingerprinting from observable evidence (headers, generator meta tags, JS framework signatures, server banners) with a confidence indicator, not a certainty claim.
- Security-indicator observation: HTTPS, HSTS, CSP, X-Frame-Options, cookie flags (Secure/HttpOnly/SameSite) – record presence/absence,

do not classify severity.

3. Output contract

- Convert everything into the shared `DiscoveryResult` Pydantic model in `app/models/schemas.py`. Do not invent your own output format.
- If a field you need doesn't exist in the schema, flag it to the team rather than silently changing the contract.

4. Backend orchestration (backend/)

- REST endpoint(s), e.g. `POST /analyze` (accepts URL + optional protection-priority list), `GET /analyze/{id}` (job status/result).
- Calls discovery module → receives `DiscoveryResult` → passes to the security-analysis pipeline (treat this as a black-box function/API call – you do not implement what's inside it) → receives `SecurityAnalysis` → stores/returns it.
- Background job handling if analysis takes more than a couple of seconds (queue, async task, or simple polling status).
- Basic error handling: invalid URL, unreachable target, analysis timeout, upstream failure.

5. Testing

- Test against your own local test apps or explicitly authorized targets only. Never test against arbitrary third-party sites.
- Validate `DiscoveryResult` output matches the schema exactly (write a schema-validation test).

DELIVERABLE FORMAT

Your final output to the rest of the team is a validated `DiscoveryResult` JSON object per the shared schema, plus working backend endpoints that the frontend can call and the security layer can consume.

If you are ever unsure whether something is in scope, default to NOT building it and ask the team instead.

MEMBER 3 — FRONTEND + PLATFORM + DEVOPS (≈20%)

PROJECT: Automated Threat Modeling & Attack Path Mapper (12-day MVP)

YOUR ROLE: Frontend + Platform + DevOps Engineer (~20% of the project)

MISSION

You build the interface that turns the `SecurityAnalysis` object (produced by the cybersecurity/AI layer, delivered via the backend) into something a human can understand at a glance and drill into for detail. You also own

deployment plumbing.

YOU OWN THESE AREAS ONLY:

- frontend/
- Deployment/DevOps config (Dockerfile(s), docker-compose, CI/CD config, env/config files, logging/monitoring setup)

YOU MUST NOT TOUCH:

- app/security/, app/rag/, app/ai/, knowledge_base/ (Cybersecurity/AI lead)
- app/discovery/, backend/ internals (Discovery/Backend member) – you may call backend endpoints, but do not reimplement backend logic in the frontend.
- app/models/schemas.py – READ ONLY. Consume SecurityAnalysis as defined here; if a field is missing for a UI you want to build, propose the addition to the team rather than inventing your own shape.

SCOPE BOUNDARY

Do not implement any cybersecurity reasoning, risk calculation, or threat classification in the frontend. If the SecurityAnalysis object says a finding is "potential" or "inferred," display it exactly that way – do not upgrade language to sound more certain (e.g., never turn "potential authentication abuse" into "site is vulnerable to brute force"). Your job is faithful, clear presentation of what the backend sends you.

WHAT YOU BUILD

1. Analysis submission UI

- URL input field + optional "protection priority" selection (e.g., authentication, data exposure, web application).
- Analyze button → calls backend POST /analyze.
- UI states: idle → submitting → analyzing (poll/await job status) → completed → error (clear message on invalid URL / unreachable target / analysis failure).

2. Results dashboard

- Overall risk badge (Low/Medium/High/Critical) as returned by the backend – never computed client-side.
- Summary counts: assets found, threats found, severity breakdown.

3. Asset view

- Card/list per asset: name, type, status (Observed/Inferred/Unknown), confidence, supporting evidence – all pulled directly from SecurityAnalysis.assets.

4. Findings / threat view

- Per finding: threat name, severity, affected asset, evidence, confidence, relevant OWASP/CWE/MITRE tags, recommended mitigation –

- all as provided by the backend.
 - Preserve hedged language ("potential," "inferred") in the UI copy.
5. Attack-path visualization
- Render the attack-path graph data (nodes/edges) supplied by the backend using a graph library of your choice (React Flow, Cytoscape.js, D3, or Graphviz rendering).
 - You only need to visualize the graph structure given to you – do not compute paths or graph logic yourself.
6. Recommendations view
- List recommendations with priority and the related threat/asset, as provided.
7. Reports
- Download/export of the analysis as PDF/JSON/CSV, generated from the SecurityAnalysis data already retrieved (client-side export or a simple backend-assisted export endpoint you build in coordination with Member 2).
8. Auth & history (if time allows in scope)
- Basic signup/login, session handling, per-user analysis history, project ownership.
9. Deployment/DevOps
- Dockerize frontend (and coordinate with Member 2 on docker-compose for the full stack).
 - Basic CI/CD (lint/build/test on push), environment config, minimal logging/monitoring for the deployed app.

DELIVERABLE FORMAT

A working UI that takes a SecurityAnalysis JSON object (matching the shared schema) and renders it faithfully – plus the Docker/CI setup needed to run and deploy the full three-person stack.

If you are ever unsure whether something requires cybersecurity logic you shouldn't own, default to NOT building it and ask the team instead.

Quick Reference: The Three Boundaries

Boundary	From	Contract	To
#1	Member 2 (discovery)	DiscoveryResult	You (security/AI lead)

#2	You (security/AI lead)	SecurityAnalysis	Member 3 (frontend)
#3	Member 3 (frontend)	UI calls	Member 2 (backend API)

Both prompts above repeatedly reinforce: **read the shared schema, don't edit it unilaterally, don't build outside your folder, escalate uncertainty to the team instead of guessing.** That's what keeps three AI-assisted workstreams from colliding over 12 days.